THE DIRECT DEDEKIND–MERTENS PROOF AND ITS UNIT FORMALISATION

This note presents the argument in the supplied *A direct proof of the Dedekind-Mertens Lemma*, dated April 5, 2006 (`mertens.pdf`), and describes its checked formalisation in Unit. The proof works over every commutative ring, without a noetherian hypothesis.

1. The subgroup theorem

Let A be a commutative ring and let

$$P = \sum_{i=0}^{n} f_i X^i, \qquad Q = \sum_{j=0}^{m} g_j X^j, \qquad PQ = \sum_{k} h_k X^k.$$

The bounds n, m are formal degrees: the displayed leading coefficients need not be nonzero. Let F, G, H be the additive subgroups of A generated by the coefficients of P, Q, PQ, respectively. For additive subgroups L, M, write LM for the additive subgroup generated by all ab with $a \in L$, $b \in M$. Set $F^0 = \mathbb{Z}1$. For $a \in A$, the notation aL denotes the subgroup of multiples $a\ell$ with $\ell \in L$.

Theorem 1. *One has $F^{m+1}G = F^m H$.*

Proof. The inclusion $F^m H \subseteq F^{m+1}G$ follows from the coefficient formula for PQ, which gives $H \subseteq FG$. We prove the other inclusion by induction on m. For $m = 0$, the coefficients of PQ are $f_i g_0$, so $FG = H$.

Suppose $m > 0$, set $Q_- = Q - g_m X^m$, and let G_- be the subgroup generated by $g_0, \dots, g_{m-1}$. Extend the notation by $f_i = 0$ outside $0 \le i \le n$. Every coefficient of PQ_- is $h_k - f_{k-m} g_m$, hence lies in $H + g_m F$. The induction hypothesis applied to P, Q_- gives

$$F^m G_- \subseteq F^{m-1} H + g_m F^m.$$

Writing $K = F^m H$ and multiplying by f_i, we obtain

$$f_i F^m G_- \subseteq K + F^m f_i g_m. \tag{1}$$

The coefficient h_{i+m} gives

$$f_i g_m = h_{i+m} - \sum_{j<m} f_{i+m-j} g_j \in H + \sum_{\ell>i} f_\ell G_-.$$

Together with (1), this implies

$$f_i F^m G_- \subseteq K + \sum_{\ell>i} f_\ell F^m G_-.$$

Descending induction on $i = n, n-1, \dots, 0$ therefore proves $f_i F^m G_- \subseteq K$ for every i, hence $F^{m+1} G_- \subseteq K$.

To include the remaining coefficient g_m, multiply the same coefficient identity by F^m. The terms $f_\ell F^m G_-$ already belong to K, so $F^m f_i g_m \subseteq K$ for every i. Thus $F^{m+1} g_m \subseteq K$ as well. Since $G = G_- + \mathbb{Z} g_m$, the required inclusion follows. □

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Corollary 2 (Dedekind–Mertens)**.** *Writing* $\mathrm{c}(P)$ *for the ideal generated by the coefficients of* P*, one has*

$$\mathrm{c}(P)^{m+1}\mathrm{c}(Q) = \mathrm{c}(P)^m\mathrm{c}(PQ).$$

Proof. Take the ideals generated by the two equal subgroups in Theorem 1. Their generators are respectively $f_{i_0}\cdots f_{i_m}g_j$ and $f_{i_1}\cdots f_{i_m}h_k$. □

Corollary 3. *If* $PQ = 0$*, then*

$$f_{i_0}\cdots f_{i_m}g_j = 0$$

for every tuple of indices. In particular, $\mathrm{c}(P)^{m+1}\mathrm{c}(Q) = 0$.

2. The coefficient order used by Unit

The polynomial library stores coefficients with the constant term first. The formal proof therefore removes the constant term of Q and uses ascending induction. This reverses the coefficient order in the preceding argument without changing its algebraic content.

Write $Q = g + XQ'$, let G' be the subgroup generated by the coefficients of Q', and put $H' = \langle\text{coefficients of } PQ'\rangle$. For $m > 0$ the outer induction hypothesis is $F^mG' \subseteq F^{m-1}H'$. Put $K = F^mH$ and consider

$$I(i): \qquad f_iF^mG' \subseteq K.$$

The coefficient identity is

$$h_i = gf_i + (PQ')_{i-1},$$

where the last term is zero for $i = 0$. If $I(\ell)$ holds for all $\ell < i$, the convolution formula for $(PQ')_{i-1}$ gives $F^m(PQ')_{i-1} \subseteq K$. It follows that

$$F^mf_ig \subseteq K. \tag{2}$$

To prove $I(i)$, multiply the outer induction inclusion by f_i. Each resulting generator has the form $f_it(PQ')_j$, where t is a product of $m-1$ coefficients of P. Now

$$f_it(PQ')_j = f_ith_{j+1} - tf_{j+1}f_ig.$$

The first term belongs to K by definition, and the second belongs to K by (2), since $tf_{j+1} \in F^m$. This proves $I(i)$. Primitive recursion supplies the finite list of earlier statements needed at each index. Finally, $I(i)$ handles the coefficients of Q', while (2) handles g.

3. What has been formalised

The development is in `unit/sag/mertens/`. It checks the entire univariate subgroup theorem, both inclusions of the ideal formula, and the coefficient-annihilation assertion of Corollary 3. The coefficient ring is arbitrary.

Membership is represented by explicit certificates. For a family (a_i), a subgroup certificate is a finite expression built from 0, the a_i, addition and negation, together with an equality identifying its value. Ideal certificates additionally allow arbitrary ring multiples of the generators. Thus the strong theorem really uses only integer signed sums. Equality of the generated subgroups or ideals is expressed by maps on certificates in both directions; no equality of the certificate types is asserted.

- `AdditiveSpan.seg` defines signed-sum certificates, their elimination into additive subgroups, and bounded induction.
- `CoefficientProducts.seg` proves the product-coefficient formula with the bounds required by the inner induction.
- `DedekindMertens.seg` contains the two inductions and the explicit certificates for $F^{m+1}G \subseteq F^mH$.

- `IdealSpan.seg` defines ideal certificates. `ContentFormula.seg` proves both subgroup inclusions and both ideal inclusions, using coefficient-product generating families directly.
- `ZeroProduct.seg` proves coefficient annihilation when PQ is zero coefficientwise.

For a coefficient list p representing P, and $Q =$ `cons` $g\ q$ with $m =$ `length` q, the main interface is

```
module D = DedekindMertens[l].Over[A]
D.theorem p q g word j
```

Here `word` has $m+1$ coefficient indices; the result is a signed-sum certificate for their coefficient product times g_j. The full formulas are available through

```
module E = ContentFormula[l].Over[A,p,g,q]
E.subgroup-forward a certificate
E.subgroup-backward a certificate
E.ideal-forward a certificate
E.ideal-backward a certificate
```

The proof uses Unit's data eliminators and equality rules. It adds no axiom and requires neither univalence or replacement parameters, choice, excluded middle, nor a noetherian hypothesis. It uses raw coefficient lists, so it does not require polynomial quotients or a decision procedure for degrees. Zero polynomials and zero leading coefficients are allowed.

From `unit/`, the checking command is

```
./segc sag/mertens all
```

The check on October 25, 2026 reported *14 modules, 0 failed*, including imported library modules.

The multivariate content lemma and its localization application in `CAR1.pdf` have not yet been connected to this development. The separate Jacobian formalisation still assumes its explicit annihilator-stabilization input; the noetherian reduction remains outside that formalisation. Those limitations concern the application, not the Dedekind–Mertens theorem proved here.